

Automated Pneumonia Screening from Chest Radiographs: A ResNet-50 Approach

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Abstract

Pneumonia remains one of the leading causes of morbidity and mortality worldwide, particularly among young children and the elderly. Timely diagnosis from chest radiographs is hindered by a global shortage of trained radiologists and the subjective nature of manual interpretation. This paper presents an automated screening system based on transfer learning with a ResNet-50 backbone to detect pneumonia from chest X-ray images. The system is trained on the publicly available Chest X-Ray Images (Pneumonia) dataset and incorporates class weighting to mitigate severe class imbalance. To enhance clinical trust, we leverage Gradient-weighted Class Activation Mapping (Grad-CAM) visualizations to highlight salient lung regions used by the model for its predictions and propose a deployment-ready decision support pipeline. Experimental results on the held-out test set yield an accuracy of 86.70%, recall of 96.67%, F_1 score of 0.90 and AUC-ROC of 0.94, illustrating how the proposed system can assist clinicians with more consistent and interpretable diagnoses.

CCS Concepts

• **Computing methodologies** → **Image classification; Transfer learning; Neural networks.**

Keywords

Pneumonia detection, Deep learning, ResNet-50, Chest X-ray, Transfer learning, Grad-CAM, Clinical decision support

1 Introduction

Pneumonia is an acute infection that inflames the air sacs in one or both lungs. Despite being preventable and treatable, it remains a major global health burden, causing millions of deaths annually. According to recent estimates, pneumonia was responsible for approximately 2.5 million deaths in 2019 [1], disproportionately affecting children under five and adults over 70 years of age. Early diagnosis is critical for guiding treatment and improving outcomes; however, expert interpretation of chest radiographs (CXR) is labour-intensive and often unavailable in resource-limited settings. Artificial intelligence (AI) offers the potential to automate screening and to provide decision support when radiologists are scarce.

In this work we develop an automated screening pipeline based on the ResNet-50 convolutional neural network (CNN) architecture [2]. Transfer learning from models pre-trained on ImageNet enables effective training on comparatively small medical datasets,

while class weighting and early stopping address class imbalance and overfitting. We also integrate Gradient-weighted Class Activation Mapping (Grad-CAM) [7] to produce heatmaps that localise regions of interest, thereby enhancing model interpretability.

Our contributions are summarised as follows:

- We review prior deep-learning approaches to pneumonia detection and identify key limitations regarding clinical deployment and interpretability.
- We present a comprehensive pipeline consisting of data pre-processing, model fine-tuning, weighted loss functions and validation strategies to train a robust ResNet-50 classifier on the Chest X-Ray Images (Pneumonia) dataset.
- We report experimental results including confusion matrices, classification metrics and receiver operating characteristic (ROC) curves, and discuss the real-world applicability of our system.
- We outline a deployment scenario for a clinical decision support system that delivers predictions and Grad-CAM explanations to healthcare professionals.

The remainder of the paper is organised as follows: Section 2 reviews related work; Section 3 formulates the problem; Section 4 describes the dataset; Section 5 details the methodology; Section 6 presents results; Section 7 discusses deployment; Section 8 outlines future work; and Section 9 concludes the paper.

2 Related Work

Deep learning has revolutionised automated interpretation of medical images. Rajpurkar et al. introduced CheXNet, a 121-layer DenseNet trained on the ChestX-ray14 dataset to perform pneumonia classification [6]. While achieving radiologist-level performance on their test set, the model was not optimised for clinical deployment and provided limited interpretability. Wang et al. curated a large-scale CXR dataset containing over 100 000 images across 14 thoracic pathologies and compared various CNN architectures [8]. However, the multi-label nature of their benchmarks diluted performance on individual diseases.

Kermay et al. collected a paediatric CXR dataset of 5 856 images and trained a transfer-learning model achieving over 92% accuracy in distinguishing normal from pneumonia cases [3]. Despite promising results, their model did not differentiate pneumonia subtypes and lacked deployment-ready explanations. More recently, Selvaraju et al. proposed Grad-CAM as a general-purpose visualisation technique for interpreting CNN predictions [7], which has since been adopted widely in medical imaging to improve clinical trust.

Table 1: Distribution of images in the Chest X-Ray dataset.

Class	Train	Validation	Test
Normal	1 171	170	234
Pneumonia	3 418	457	390
Total	4 589	627	624

Our work builds on these efforts by combining a ResNet-50 backbone with class weighting and Grad-CAM visualisation to produce a transparent, resource-efficient screening system suitable for deployment in resource-constrained settings.

3 Problem Identification

The burden of pneumonia falls heaviest on paediatric and elderly populations. In many low-resource regions radiologists are scarce; healthcare workers must manage heavy workloads and depend on delayed referrals, leading to treatment gaps. The unmet needs include reliable and fast pre-screening tools, consistent interpretations across practitioners and the ability to prioritise cases that require urgent attention. Our proposed system addresses these gaps by providing a 24/7 decision support tool that can be integrated into existing clinical workflows, thereby reducing diagnostic delays and relieving overworked radiologists.

4 Dataset

We utilise the Chest X-Ray Images (Pneumonia) dataset curated by Kermany et al. [3] and made publicly available by Mooney [5]. The dataset contains 5 856 JPEG CXR images sourced from Guangzhou Women and Children’s Medical Centre and labelled as either Normal or Pneumonia. Following standard practice, we create a validation split by allocating 15% of the training data to validation. Table 1 summarises the distribution across training, validation and test sets.

The dataset is appropriate because it originates from clinical settings and has been used in peer-reviewed studies [3]. Its size is sufficient for transfer learning and serves as a widely adopted benchmark in the literature. We note that the class imbalance ratio is approximately 2.9:1 (Pneumonia:Normal), which we address through weighted loss functions in Section 5.3.

5 Methodology

Our pipeline consists of image preprocessing, model architecture adaptation, training strategy and interpretability. We briefly describe each component.

5.1 Data Preprocessing

All images are resized to 224×224 pixels to match the ResNet-50 input requirements. Because the original CXRs are greyscale, we convert them to three-channel RGB by duplicating the single channel, allowing us to leverage pre-trained weights. Images are then converted to PyTorch tensors and normalised using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). Table 2 lists the preprocessing stages.

Table 2: Image preprocessing pipeline.

Stage	Operation	Purpose
1	Resize	Rescale variable-sized CXRs to 224×224 for ResNet input.
2	RGB conversion	Duplicate the greyscale channel to form a three-channel image for compatibility with ImageNet pre-training.
3	Tensor conversion	Map pixel intensities from [0, 255] to [0, 1] floating-point tensors.
4	Normalisation	Standardise pixel distributions with ImageNet mean and standard deviation to accelerate convergence.



Figure 1: High-level overview of the ResNet-50 architecture used in this work. The final fully connected layer is replaced for binary classification.

5.2 Model Architecture

We employ ResNet-50 [2] as the backbone due to its favourable trade-off between depth and parameter count. ResNet’s identity skip connections mitigate vanishing gradients and enable effective training of deep networks. We replace the final fully connected layer (originally 1 000-class ImageNet output) with a new linear layer that outputs two logits representing Normal and Pneumonia classes. Figure 1 depicts the high-level architecture.

5.3 Training Strategy

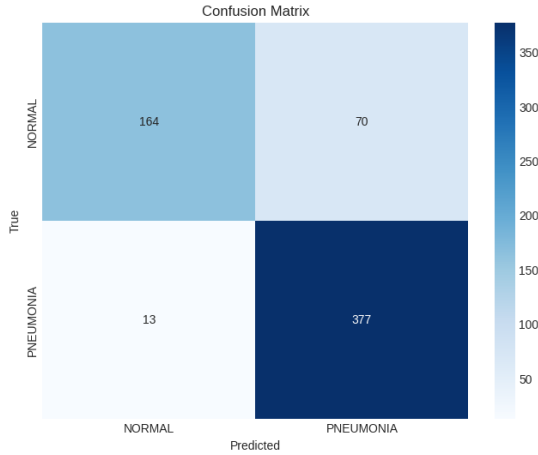
The model is initialised with ImageNet pre-trained weights. All backbone layers are frozen and only the replacement fully connected layer is trained, constituting a feature-extraction transfer-learning approach. We use cross-entropy loss with class weights inversely proportional to class frequencies to compensate for class imbalance, assigning higher weight to the minority Normal class. The Adam optimiser [4] is configured with a learning rate of 1×10^{-2} , batch size of 32 and default momentum parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$). Training is performed for up to 10 epochs with early stopping triggered if the validation loss fails to improve for three consecutive epochs (patience = 3). The model checkpoint with the lowest validation loss is saved as the final model.

5.4 Interpretability

To enhance clinical trust, we employ Grad-CAM [7] to generate localisation heatmaps indicating the regions that contribute most to the classification decision. Specifically, we hook into the final convolutional block (layer4[-1]) of ResNet-50, compute the gradient of the target class score with respect to the feature maps, and produce a weighted combination that is upsampled and overlaid

Table 3: Test set performance metrics.

Class	Precision	Recall	F ₁	Support
Normal	0.93	0.70	0.80	234
Pneumonia	0.84	0.97	0.90	390
Accuracy		0.87		624
Macro avg	0.88	0.83	0.85	624
Weighted avg	0.87	0.87	0.86	624

**Figure 2: Confusion matrix on the test set. The model achieves high sensitivity (96.67%) with a modest false-positive rate.**

on the original CXR. By providing these visual explanations, clinicians can confirm that the model focuses on lung fields and opacity regions, improving acceptance of the AI tool.

6 Results

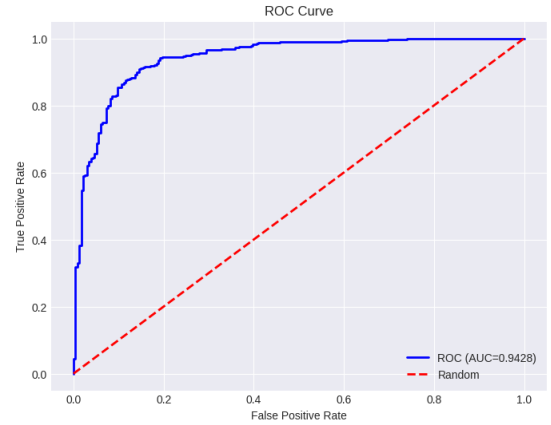
We evaluate the trained model on the held-out test set comprising 624 images (234 Normal, 390 Pneumonia).

6.1 Quantitative Results

Table 3 summarises the key performance metrics. The model achieves an overall accuracy of 86.70% with a recall (sensitivity) of 96.67% for the pneumonia class, which is clinically desirable as it minimises missed diagnoses. The AUC-ROC of 0.94 indicates strong discriminative ability across classification thresholds.

Figure 2 shows the confusion matrix, revealing 164 true negatives, 377 true positives, 70 false positives and 13 false negatives. The low false-negative count (13 out of 390) confirms that the weighted loss successfully prioritises sensitivity, a critical property for a screening tool.

Figure 3 presents the ROC curve with an AUC of 0.94, demonstrating that the model maintains a favourable trade-off between sensitivity and specificity across a range of decision thresholds.

**Figure 3: Receiver operating characteristic (ROC) curve demonstrating the trade-off between sensitivity and specificity (AUC = 0.94).****Figure 4: Training and validation loss and accuracy across epochs. Early stopping is triggered after minimal improvement.**

6.2 Training Dynamics

Figure 4 shows the training and validation loss and accuracy across epochs. The model converges rapidly due to the frozen backbone, with early stopping triggered after epoch 5 when validation loss plateaus.

6.3 Qualitative Analysis

Grad-CAM attention maps confirm that the model attends to anatomically plausible regions. Figure 5 shows representative heatmap overlays for correctly classified images, where the model focuses on pulmonary fields and opacity regions. Error analysis reveals that misclassifications often correspond to borderline cases with subtle infiltrates or atypical positioning.

7 Real-World Application

We envision integrating the proposed system into a clinical decision support workflow. Patient CXRs are uploaded to a cloud or edge server hosting the ResNet-50 model. The server returns class probabilities and Grad-CAM visualisations via a secure API. Radiologists can use the tool as a second opinion, rapidly triaging normal cases and flagging likely pneumonia for further review.

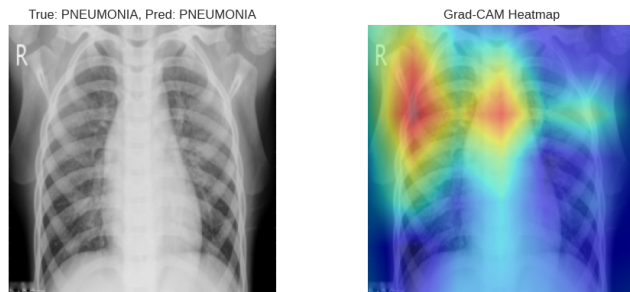


Figure 5: Grad-CAM heatmap overlay on correctly classified pneumonia cases, showing the model attends to relevant lung opacity regions.

For settings with limited connectivity, the model can be deployed on an edge device attached to the X-ray machine to provide instant results without requiring a remote server. Key considerations for deployment include data privacy, regulatory compliance (such as adherence to medical device regulations) and continuous monitoring for demographic biases and dataset drift.

8 Future Work

To evolve this prototype into a clinically viable software as a medical device (SaMD), future work will focus on:

- **Data diversity:** Augmenting the dataset with images from diverse demographics and adult populations to mitigate demographic bias, as the current dataset is restricted to paediatric patients from a single centre.
- **Fine-tuning the full network:** Unfreezing deeper layers with a discriminative learning rate schedule to improve feature adaptation beyond ImageNet representations.
- **Multi-class classification:** Extending the model to distinguish bacterial, viral and atypical pneumonia sub-types and to provide severity grading.
- **Data augmentation:** Incorporating random rotations, horizontal flips and intensity jittering to improve generalisation and robustness.
- **Ensemble methods:** Combining multiple architectures (e.g., DenseNet, EfficientNet) to improve robustness and calibration.
- **Regulatory compliance:** Ensuring adherence to medical device regulations and ethical considerations such as fairness and explainability.

9 Conclusion

We have presented an automated pneumonia screening system leveraging transfer learning with ResNet-50 and interpretability via Grad-CAM. The system effectively classifies chest X-rays into normal and pneumonia classes, achieving 86.70% accuracy, 96.67% recall and an AUC-ROC of 0.94 on the benchmark dataset. By addressing class imbalance through weighted loss functions, incorporating early stopping and providing visual explanations, the model demonstrates potential for deployment as a decision support tool in resource-constrained healthcare environments. Ongoing work will

expand the dataset, explore full network fine-tuning and extend the model to more granular classification tasks, paving the way for clinically useful AI-driven diagnostic aids.

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